DASC 3240, L08

Barplot and more – Assignments

2023/24 SPRING SEMESTER

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Today's outline

- Barplot and more – Assignments
  - Data
  - Assignments

Data

Datasets

Rdatasets 1.0.0 Available datasets

What is this?

`Rdatasets` is a collection of 2264 datasets which were originally distributed alongside the statistical software environment `R` and some of its add-on packages. The goal is to make these data more broadly accessible for teaching and statistical software development.

What is included?

The list of available datasets (csv and docs) is available here:

- HTML index
- CSV index

On the github repository you will also find the scripts I use to scrape data and update the website.

https://vincentarelbundock.github.io/Rdatasets/

ChickWeight

https://vincentarelbundock.github.io/Rdatasets/doc/datasets/ChickWeight.html
https://vincentarelbundock.github.io/Rdatasets/doc/datasets/ChickWeight.csv

Weight versus age of chicks on different diets

Description

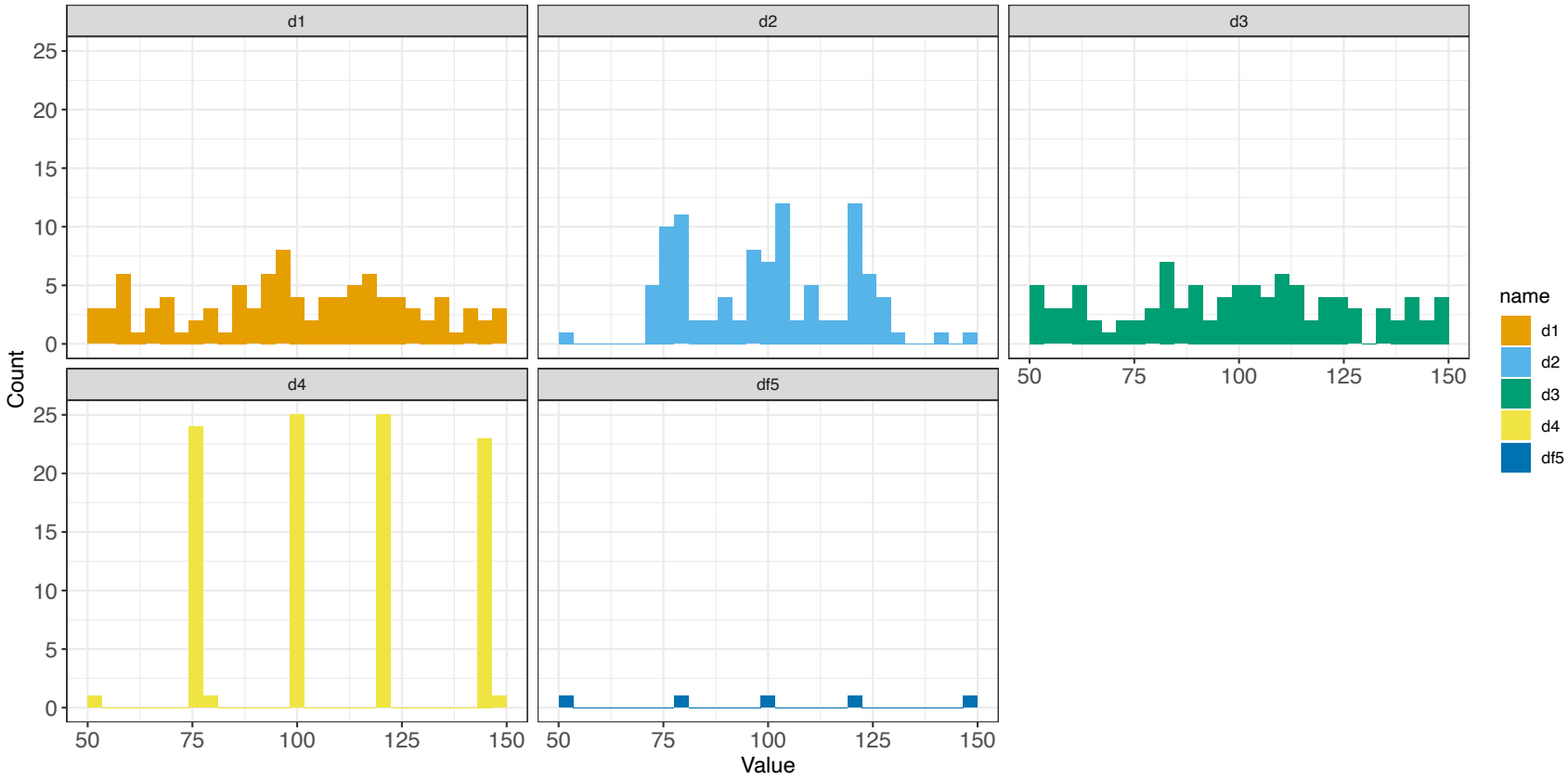
The ChickWeight data frame has 578 rows and 4 columns from an experiment on the effect of diet on early growth of chicks.

	weight	Time	Chick	Diet
1	42	0	1	1
2	51	2	1	1
3	59	4	1	1
4	64	6	1	1
5	76	8	1	1
6	93	10	1	1

weight the body weight of the chick (gm).
Time the number of days since birth when the measurement was made.
Chick levels 18 < ... < 48 giving a unique identifier for the chick. The ordering of the levels groups chicks on the same diet together and orders them according to their final weight (lightest to heaviest) within diet.
Diet levels 1, ..., 4 indicating which experimental diet the chick received.

Custom datasets

The datasets are created by myself to show how using barplots and boxplots might be misleading.



https://github.com/ong8181/DASC3240/blob/main/data/d1.csv ~ d5.csv

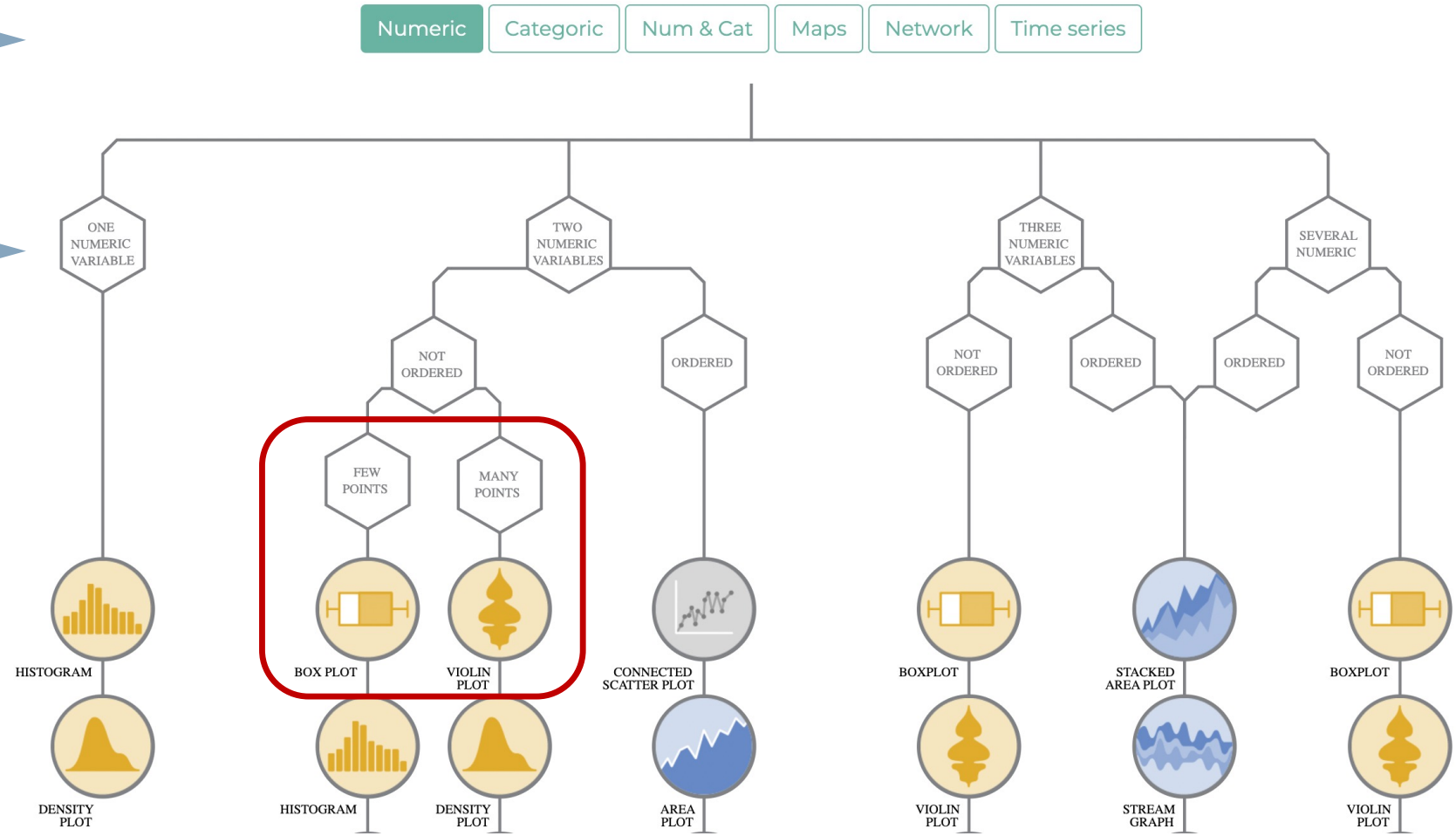
Assignments

BARPLOTS AND MORE

How many variables? Numeric or categorical?

Numeric or categorical?

The number of variables



https://www.data-to-viz.com/

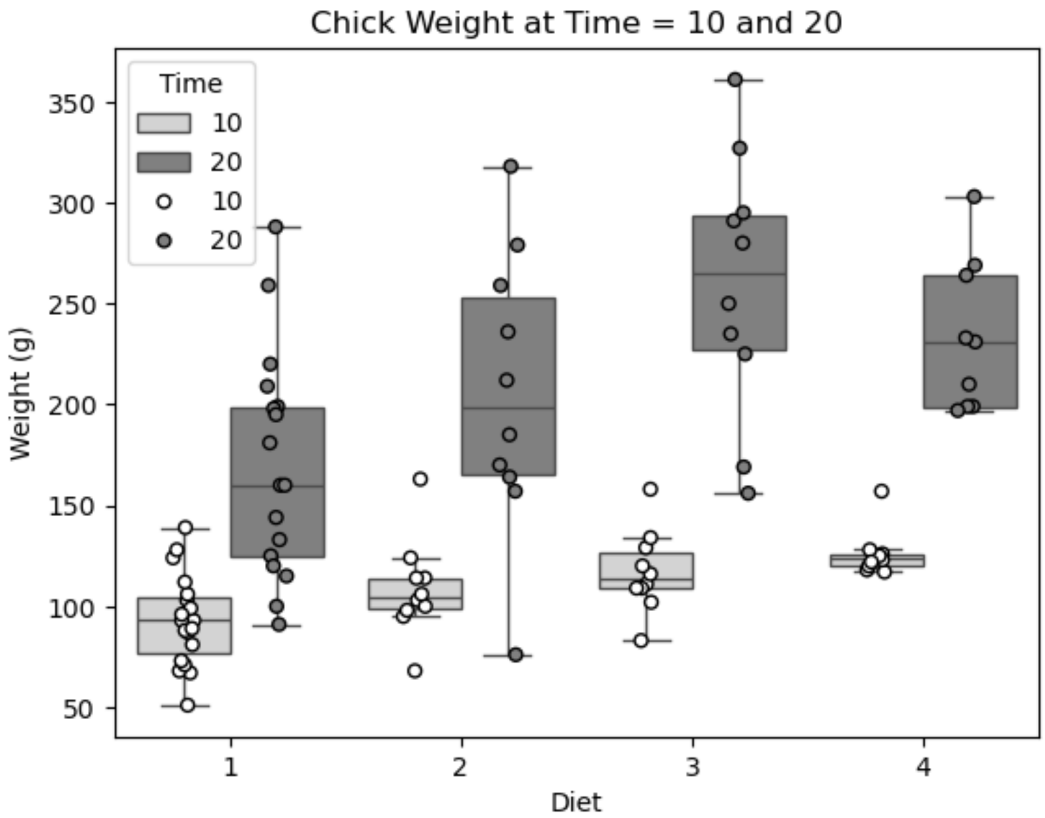
Assignments

- **ChickWeight**
  1. Extract data with **Time = 10** and visualize the effect of **Diet** on **weight** using boxplots overlaid with jitter plots (please be careful of outlier points).
  2. Extract data with **Time = 20** and visualize the effect of **Diet** on **weight** using boxplots overlaid with jitter plots (please be careful of outlier points).
  3. Combine the two figures generated in **Task 1** and **Task 2** so that readers can easily understand the effects of **Diet** and **Time** on **weight** (an example is provided).
- **Custom datasets**
  1. Combine the custom datasets (**d1**, **d2**, **d3**, **d4**, and **d5**) into one 'pandas.DataFrame'. Visualize them using boxplots overlaied with beeswarm plots (an example is provided).

Assignments

- **ChickWeight**

  3. Combine the two figures generated in **Task 1** and **Task 2** so that readers can easily understand the effects of **Diet** and **Time** on **weight** (an example is provided).



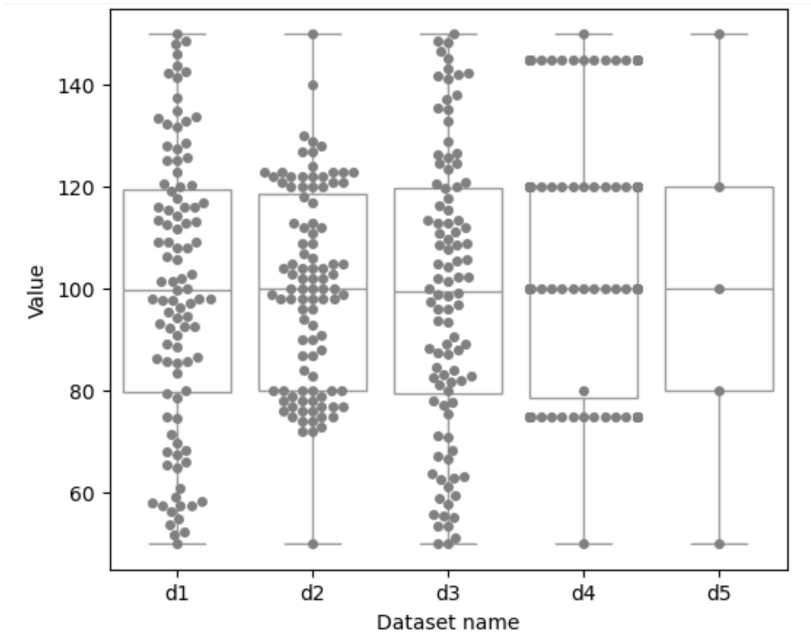
Assignments

- **Custom datasets**

1. Combine the custom datasets into one ‘pandas.DataFrame’. Visualize them using boxplots overlaied with beeswarm plots (an example is provided).

Tidy-data

	id	value	dataset
0	1	150.000000	d1
1	2	125.302795	d1
2	3	147.974703	d1
3	4	122.979472	d1
4	5	128.737786	d1
...	...	...	...
0	1	150.000000	d5
1	2	120.000000	d5
2	3	100.000000	d5
3	4	80.000000	d5
4	5	50.000000	d5



https://cran.r-project.org/web/packages/tidyr/vignettes/tidy-data.html